Linear Models

Pre-Lab

Purpose: In this lab you will learn how to model a data set which appears to be linear (after bit of a fix, anyway!), and use these ideas to model the velocity of a falling book.

Preview: How can you tell (visually and algebraically) if data would be well-modeled by a linear function? If a linear function is appropriate, how can you find an equation for a linear function that will be a good model for the data? How good is the model you obtain? In the course of finding linear models you will encounter the idea of the average rate of change of a function.

Part I: Height

Suppose you drop an object a short distance and measure its distance from the floor at various times during its fall. Will the object fall at a constant rate? Can the distance data be well-modeled by a linear function? Let's find out.

Below is a data set recorded from an experiment where a book was dropped from a height of about 1.542 meters. The table gives the height at every .03 second interval.

t	0	.060	.090	.120	.150	.180	.210	.240	.270	.300
s	1.542	1.527	1.498	1.468	1.436	1.379	1.329	1.258	1.195	1.111
t	.330	.360	.390	.420	.450	.480	.510	.540	.570	.600
s	1.023	.924	.806	.694	.745	.646	.664	.623	0.00	.651

1. Open the spreadsheet for this lab and make a copy, renaming it with your name. You will find the data from the table above in Columns A and B.

2. Make a scatter plot of this data (be sure to get the best window possible). Your plot should resemble half of a parabola – at least in part.

3. **Working with experimental data:** Do some points seem incorrect? What could explain the discrepancies? Edit your chart to exclude these points from it. Your plot should now more closely resemble half a parabola.

Part II: Motion and Rates of Change

4. **Examining the data:**

   (a) Compute the *average rate of change of height* over the first time interval by hand. To do this, take the difference between the first two heights, and divide it by the different between the first two times. If you do this right, you should get -0.25.

   (b) What are the units of your average rate of change (hint: you divided height by time!)? What does this average rate of change of height represent in terms of the motion of the falling book?

   Note: We will use the number that you just got as an estimate of the velocity in the *center of the first time interval.* That is, at 0.03 seconds.

   (c) In cell D2, insert a formula that computes the center of the first time interval (i.e., the average of the first two times). Fill this formula down to compute the centers of all subsequent intervals until the the middle of the last interval shown on your modified chart. If you do this right, your last entry should be 0.405 seconds.

   (d) In column E, insert a formula that computes the average rate of change of the height over each time interval. Fill this formula down to compute the average rates of change for all subsequent intervals until the last interval shown on your modified chart. If you do this right, your last entry should be about -3.7.

   (e) Lastly, insert of a scatter plot of the average rates of change of height vs. the centers of the time intervals.

5. Go do the short Gradescope assignment for this pre-lab.

Main lab

At the start of lab, one of you should share their sheet with the rest of group, and you should rename it with all your names. Also, have a shared Google Doc for your notes.

6. Consider the plot of the *rates of change of height*. What do you notice about your average rates of change? Do they trend upward or downward (in other words, overall, are they increasing or decreasing)? What does that mean in real world terms?

7. Now consider the plot of the *heights of the object*. What characteristic(s) of the height graph indicate that the book was dropped rather than thrown downward

Re-examining the Average Rates of Change

This part of the lab will focus on the scatter plot of *velocities (average rates of change)* you created at the end of the pre-lab. From that plot, what do you observe about the velocity of the falling book?

Modeling the velocity: Find a linear model $v = mt + b$ for the velocity data by using the Linear Trendline feature in your spreadsheet. Make sure you display the formula on the chart.

Interpreting the model:
The key idea here to understand the difference between the signs of the velocities and the trend (or slope) of the plot.

8. What does the slope m of the *velocity* graph represent in this case? (Hint: What are the units of the average rate of change of the average rates of change of the distances?)

9. What is the meaning of the fact that your data for v are negative?

10. What is the practical meaning of the fact that your model of the velocity has a negative slope?

11. Near the earth's surface the acceleration of gravity is approximately -9.8 meters/second per second. State at least two reasons why the estimate of the acceleration due to gravity from your trendline differs from this.

12. According to your model, what was the velocity of the book at t = 0? Does this result agree with what the researchers stated occurred during the experiment? If not, explain the discrepancy.

13. According to your model, how long would it take the book to reach a velocity of -50 meters per second ($\approx$ 110 miles per hour)? Does the book ever reach this velocity? Explain.

Linear Models Lab Report Form

Answer each of the following equations using one or two complete sentences.

1. What does the slope m of the *velocity* graph represent in this case?
2. Explain why, in this situation, it makes sense that your data for v are negative.
3. What is the practical meaning of the fact that your model for velocity has a negative slope? (Note: this is a different question than the previous one!)
4. According to the linear trendline model, what was the velocity of the falling book at $t = 0$? In the experiment conducted that produced the data set you used in the lab, the people conducting the experiment reported that they dropped the book. What would this mean in terms of the velocity of the book at $t = 0$? How might you explain this discrepancy?
5. Assuming the height measurement is imperfect, explain why it is impossible to verify that the book was in fact dropped using measured height data. Specifically, if the second measured height is less than the first, explain why it is possible for the book to have been thrown upward, dropped, or thrown downward.